Tracking charged particles through matter and fields

Model comparison, Kalman filter and vertex reconstruction, in 2D

PHY571 Numerical Physics project · team of 3 · 6 weeks · Python (NumPy/SciPy; PyTorch only for the ML part)

Version 4

The project in one paragraph

We follow charged particles through a toy detector: thin silicon planes that record noisy positions (*hits*), a block of matter, and a magnet that bends the particles so that we can measure their momentum. We simulate everything ourselves (so we know the truth), then reconstruct the trajectories from the hits only. The project has two scenarios that share *one* code base:

- **Proton mode** (medical physics): 200 MeV protons cross a water phantom (a fake patient), lose energy and scatter. We reconstruct their path and exit momentum.
- **Di-muon mode** (particle physics): a J/ψ produced in a thin target decays into $\mu^+\mu^-$. We reconstruct both tracks, their common **vertex** and the **invariant mass** of the pair.

The common thread is **comparing models**: several magnets, several phantoms, several propagators, several physics models and several fitting methods, always against the same Monte Carlo truth.

Research questions.

Q1. How do multiple scattering and energy loss degrade the reconstruction of the proton path and exit momentum, and in which regime does a Kalman filter outperform a global fit?

Q2. How much does each modelling choice (field model, propagator, energy-loss and scattering model, phantom composition) change the result, and what happens when the model inside the fit differs from the one that generated the data?

Q3. How well can we reconstruct a two-track decay ($J/\psi \to \mu^+\mu^-$): vertex position and invariant-mass resolution, and how much does a vertex constraint help?

Why it matters. This is how tracks are reconstructed at LHCb, ATLAS and CMS, and how proton radiography / proton CT estimates the most likely path of each proton inside a patient. One code, two fields.

What changed since version 3

- **Strictly 2D.** We work only in the bending plane xz. The 3D extension is dropped (§1.2).
- **Model comparison is now the core** of the project (§2, §3.2, §5): 4 magnet models, 3 phantoms, up to 6 propagators, alternative energy-loss and scattering models.
- **New scenario:** $J/\psi \to \mu^+\mu^-$ with vertex fit and invariant mass (§2.7, §3.5).
- **Machine learning** as a method to compare against the classical ones, not as decoration (§4).
- Team of 3: new split of the work (§8).

1 The experimental setup

1.1 Two scenarios, one detector concept

Both scenarios use the same ingredients (silicon planes, optional block of matter, dipole magnet) and the same code. Only the geometry and the source of particles change, and both are read from a configuration file.

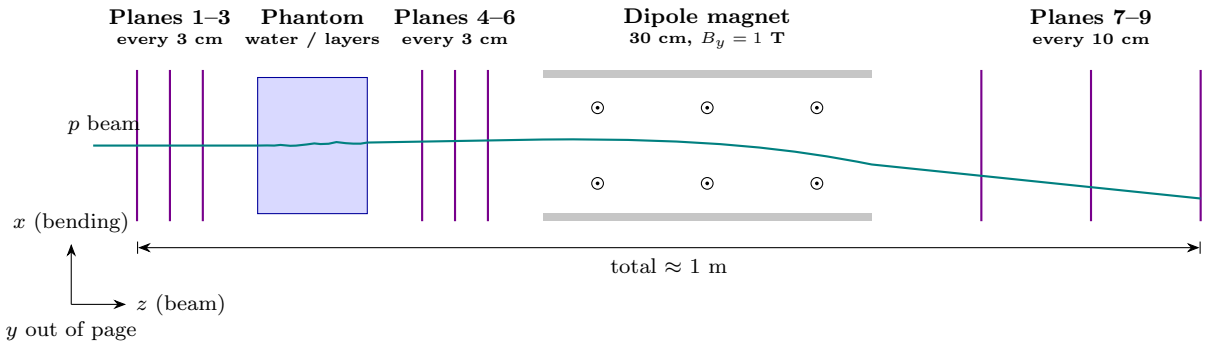


Figure 1: Proton mode, top view (plane xz). Transverse scale exaggerated. Violet: silicon planes. Circles with a dot: field B_y out of the page. The wiggles inside the phantom represent multiple scattering. Schematic, not a real detector.

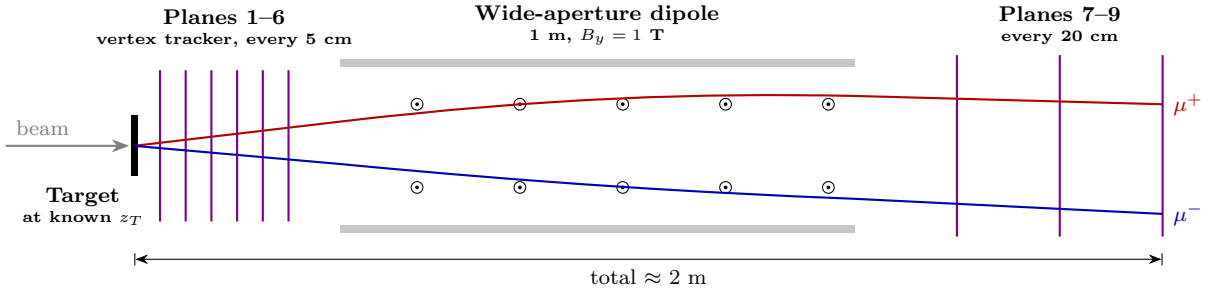


Figure 2: Di-muon mode, top view. A J/ψ produced in the thin target decays immediately into $\mu^+\mu^-$. The two muons bend in opposite directions in the magnet. No phantom. Schematic.

1.2 Coordinates, and why we stay in 2D

- z: beam direction. We use z as the "time" variable: particles are followed plane by plane in z.
- x: horizontal, transverse. The magnet bends in x. **This is the only transverse coordinate we simulate.**
- y: along the magnetic field. Nothing new happens in y: for small angles, scattering in y is an independent statistical copy of scattering in x, and the field does not bend in y.

The track state has three numbers and a 3×3 covariance:

$$\mathbf{s}(z) = \left(x,\ t_x = \tfrac{\mathrm{d}x}{\mathrm{d}z},\ q/p\right)^\top, \qquad C \in \mathbb{R}^{3\times 3}.$$

Decision: we do not go to 3D. It would multiply the bookkeeping (5×5 matrices, stereo planes) without adding a new idea to any of our research questions. For the di-muon mode, 2D has three consequences that we state openly in the report (§2.7).

1.3 What each part does

Proton mode.

Planes 1–3 measure where the proton enters and in which direction.

Phantom water or several layers (the "patient"). The proton loses energy and scatters inside it.

Planes 4–6 measure where the proton leaves the phantom and in which direction.

Magnet bends the proton. The bending angle depends on its momentum.

Planes 7–9 measure the direction after the magnet. Compared with planes 4–6 this gives the exit momentum.

Di-muon mode.

Target thin foil at a known position z_T where the J/ψ is produced. It decays on the spot ($c\tau \sim 10^{-12}$ m).

Planes 1–6 vertex tracker: measure both muon directions close to the target, to locate the vertex.

Magnet wide aperture, so that both muons fit inside. It gives each muon's momentum.

Planes 7–9 directions after the magnet, spaced 20 cm for a long lever arm.

1.4 Default parameters (fixed in week 1, all in the config)

Quantity	Proton mode	Di-muon mode
Source	protons, $T_0 = 200$ MeV, spread $\sim 1\%$	J/ψ, $p = 20$ GeV/c along z, from target at $z_T = 0$
Silicon planes	9, 300 µm, $\sigma \approx 100$ µm	9, same planes
Matter	phantom, 10 cm along z (water or layers)	none (thin target only)
Magnet	30 cm long, gap 10 cm, $B_y = 1$ T	1 m long, gap 1 m, $B_y = 1$ T
Plane positions	1–3 every 3 cm; 4–6 every 3 cm; 7–9 every 10 cm	1–6 at 5–30 cm; magnet 40–140 cm; 7–9 at 160, 180, 200 cm

Sanity numbers, proton mode

After 10 cm of water the proton leaves with $T \approx 150$ MeV, i.e. $p \approx 0.56$ GeV/c. In 1 T the bending radius is $r = p/(0.3B) \approx 1.9$ m, so in 30 cm of magnet it turns by $\approx 0.3/1.9 \approx 9°$ and reaches $x \approx 7$ cm at the last plane. **Regime: scattering-dominated.** If you change energy, phantom or field, redo this check.

Sanity numbers, di-muon mode

In the J/ψ rest frame each muon has $p^* = \sqrt{M^2/4 - m_\mu^2} \approx 1.545$ GeV/c. For $p_{J/\psi} = 20$ GeV/c the muons have $p \approx 3$–17 GeV/c (median ≈ 10) and angles up to ~ 0.15 rad, hence the wide magnet. Both muons are inside the detector in $\approx 50\%$ of the decays (toy estimate). The magnet gives each muon a kick $\Delta p_x = 0.3\,BL \approx 0.3$ GeV/c, i.e. ≈ 30 mrad at 10 GeV/c. The planes measure each direction to ≈ 0.5 mrad, so $\sigma_p/p \approx 2\%$ and σ_m of order 40 MeV before the vertex constraint. Scattering in one silicon plane is only $\theta_0 \approx 0.06$ mrad. **Regime: measurement-dominated**, the opposite of the proton mode.

2 The physics, one effect at a time, with models to compare

Each physical ingredient now comes in *several models*. Each model is a small function with the same interface, chosen by name in the config. Models are labelled (M-a, P-b, ...) so that figures

and the report can refer to them.

2.1 Motion in the magnetic field

With z as independent variable and $\kappa = 0.2998\,\mathrm{GeV\,T^{-1}m^{-1}}$, in 2D:

$$\frac{\mathrm{d}x}{\mathrm{d}z} = t_x, \qquad \frac{\mathrm{d}t_x}{\mathrm{d}z} = -\kappa\,\frac{q}{p}\left(1+t_x^2\right)^{3/2} B_y(x,z). \tag{1}$$

The slope changes faster when the field is stronger and the momentum smaller. In a uniform field the exact solution is a circle of radius $r[\mathrm{m}] = p[\mathrm{GeV}/c]/(0.3\,B[\mathrm{T}])$.

The exact-circle test is not optional

The same propagator generates the truth and feeds the filter. A wrong factor (for example $(1+t_x^2)^{1/2}$ instead of $(1+t_x^2)^{3/2}$) would be invisible when comparing them. Only the comparison with the exact circle catches it.

2.2 Magnet models

We only need B_y in the bending plane ($y = 0$). Four models, from simplest to most realistic:

Label	Model	Description
M-a	Hard edge	$B_y = B_0$ inside, 0 outside. Exact circular solution.
M-b	Enge fringe	smooth entrance and exit, $B_y(z) = B_0/\left(1+e^{P(s)}\right)$ with s the distance to the edge over the gap and P a short polynomial. Tends to M-a when the fringe length $\to 0$.
M-c	Biot–Savart	field of two finite current plates (with their return paths, the circuit must be closed), integrated numerically and evaluated at $y = 0$.
M-d	Field map	M-c tabulated on a grid, then interpolated (bilinear vs. cubic spline). This is what real experiments do.

Checks: M-c tends to $B_y = \mu_0 K$ for infinite plates. $\nabla\cdot\mathbf{B} = 0$ and $\nabla\times\mathbf{B} = 0$ outside the conductors, evaluated numerically with small steps in y (only in the test, the tracking stays in 2D). For M-d: interpolation error vs. grid spacing.

Comparison: momentum bias when the data are generated with M-c but the fit assumes M-a or M-b. Does the fringe field matter? For which magnet length?

2.3 Phantom models (proton mode)

Label	Phantom	Description
P-a	Homogeneous water	10 cm of water. WEPL equals the thickness: a clean check.
P-b	Layers along z	slabs, e.g. water / bone / lung-like / water. Material depends on z only.
P-c	Insert in x	water with a bone insert covering part of the x range. WEPL now depends on x: a 1D proton radiograph.

Each material needs ρ, Z/A, I and X_0, taken from the PDG or NIST tables (water: $Z/A = 0.555$, $I = 75\,\mathrm{eV}$, $X_0 = 36.1\,\mathrm{cm}$). At an interface the propagator must stop exactly at the boundary, never step across it.

2.4 Energy-loss models

Mean energy loss (Bethe–Bloch):

$$-\frac{\mathrm{d}E}{\mathrm{d}s} = K\,\frac{Z}{A}\,\rho\,\frac{1}{\beta^2}\left[\frac{1}{2}\ln\frac{2m_e c^2\beta^2\gamma^2 T_{\max}}{I^2} - \beta^2\right], \tag{2}$$

with $K = 0.307\,\mathrm{MeV\,mol^{-1}cm^2}$ and $\mathrm{d}s = \sqrt{1+t_x^2}\,\mathrm{d}z$. It enters the state as a third equation integrated with (1):

$$\frac{\mathrm{d}(q/p)}{\mathrm{d}z} = -\frac{(q/p)^2}{\beta}\,\frac{\mathrm{d}E}{\mathrm{d}s}\,\sqrt{1+t_x^2}.$$

Label	Model	Description
E-a	Constant $\mathrm{d}E/\mathrm{d}z$	simplest fallback; misses the Bragg peak.
E-b	Bethe–Bloch	equation (2), mean loss only.
E-c	Bethe–Bloch + straggling	adds Gaussian (Bohr) fluctuations, $\sigma_E^2 = K\,m_e c^2\,\rho\,\frac{Z}{A}\,\Delta s\,\frac{1-\beta^2/2}{1-\beta^2}$, in the simulation only.

Checks: integrated range vs. NIST PSTAR (E-b), range spread (E-c). In the di-muon mode energy loss in silicon is $\sim 0.1\,\mathrm{MeV}$ per plane and is neglected.

2.5 Multiple-scattering models

Highland formula:

$$\theta_0 = \frac{13.6\,\mathrm{MeV}}{\beta c p}\sqrt{\frac{\ell}{X_0}}\left[1 + 0.038\ln\frac{\ell}{X_0}\right]. \tag{3}$$

Label	Model	Description
S-a	Highland	Gaussian kick per slice, $\delta t_x \sim \mathcal{N}(0, \theta_0^2(1+t_x^2)^2)$.
S-b	Lynch–Dahl	improved Gaussian width, better for thick layers and mixtures.
S-c	Gaussian core + tail	a small fraction of large single-scattering kicks (Molière-like tails). Simulation only: the Kalman filter cannot represent it.

In the filter, scattering enters as the process noise Q.

Same physics in both places, or a controlled mismatch

By default the simulation and the filter use the same model. Any difference must be *deliberate*: that is the mismatch study of §5. An accidental mismatch makes the pulls wrong and you will not know whether the filter or the model is to blame.

2.6 The detector

Each plane measures x with Gaussian noise, $m_k = x_{\mathrm{true}}(z_k) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, and scatters the particle a little (thin silicon, included in both modes).

2.7 The $J/\psi \to \mu^+\mu^-$ source (di-muon mode)

Generation. The J/ψ ($M = 3.0969\,\mathrm{GeV}/c^2$) is produced at the target with momentum p along z (optionally a small angular spread). In its rest frame the two muons go back to back with

$$p^* = \sqrt{\tfrac{M^2}{4} - m_\mu^2}, \qquad E^* = \tfrac{M}{2},$$

at an angle ϕ^* drawn uniformly in $[0, 2\pi)$ (the 2D version of an isotropic decay). Boosting along z with $\gamma = E/M$, $\beta = p/E$:

$$p_x^\pm = \pm p^* \sin\phi^*, \qquad p_z^\pm = \gamma\left(\pm p^* \cos\phi^* + \beta E^*\right), \qquad t_x^\pm = p_x^\pm / p_z^\pm.$$

Both muons are then propagated with exactly the same code as the protons.

Prompt vs. displaced. By default the decay is prompt: the vertex is at z_T. As a variant, the J/ψ can come from a B-meson decay, which flies an exponential distance with mean $\beta\gamma c\tau_B$ ($c\tau_B \approx 0.5\,\text{mm}$) before decaying. Then z_v is unknown. This matters a lot in 2D (§3.5).

Why the J/ψ is a perfect test. Its natural width is $\Gamma \approx 93\,\text{keV}$, far below our resolution. The width of the reconstructed mass peak is therefore *exactly* our mass resolution.

What 2D changes for the di-muon mode (stated in the report)

1. **The kinematics is exact.** Both muons live in the xz plane, so the invariant mass is fully determined by $(t_x, q/p)$ of each track.
2. **The angular distribution is a 2D one** (ϕ^* uniform instead of $\cos\theta^*$ uniform). Irrelevant for resolution studies.
3. **The resolution is slightly optimistic**, because errors in y and t_y do not exist.

3 From hits back to the track

3.1 The game: truth vs. reconstruction

1. **Simulate (Monte Carlo).** Generate a particle (or a J/ψ), propagate everything with random scattering and energy loss, record the true state at each plane.
2. **Measure.** Add Gaussian noise to the true x at each plane. Throw away the truth.
3. **Reconstruct.** From the hits only, estimate the state at each plane (and the vertex and mass in di-muon mode).
4. **Compare with the truth.** Repeat for 10^4–10^5 events and look at the statistics.

3.2 The propagator, and the propagator models

Given the state at plane k, return the state at plane $k+1$ and the Jacobian $F_k = \partial \mathbf{s}_{k+1}/\partial \mathbf{s}_k$. Without field and energy loss, F is the drift matrix

$$F = \begin{pmatrix} 1 & \Delta z & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Label	Propagator	What we expect
R-a	Euler	first order; the bad reference that shows why we need better.
R-b	RK4, fixed step	fourth order, error $\propto h^4$. The workhorse.
R-c	RK45 (Dormand–Prince)	adaptive step with error control; few steps in the empty regions, many in the fringe field.
R-d	Runge–Kutta–Nyström	exploits that (1) is second order in x; fewer field evaluations per step (used in ATLAS).
R-e	Analytic circle	exact, uniform field only. The reference for R-a to R-d.
R-f	Neural network (ML)	see §4.

Jacobian, three ways to compare: finite differences, integrating the variational equations alongside the trajectory, and automatic differentiation (for R-f).

Figures of merit: error vs. number of field evaluations, CPU time per track, and the effect on the fit (pulls, χ^2). A propagator that is "accurate enough" is one whose error is far below σ and below the scattering.

3.3 Method 1: global least-squares fit

Write all hits as a function of the initial state and minimise

$$\chi^2 = \big(\mathbf{m} - \mathbf{h}(\mathbf{s}_0)\big)^\top V^{-1} \big(\mathbf{m} - \mathbf{h}(\mathbf{s}_0)\big).$$

Scattering makes downstream hits correlated, so V is a full 9×9 matrix; inverting it costs $O(N^3)$. The version with diagonal V (scattering ignored) is the *naive global fit*.

3.4 Method 2: Kalman filter + RTS smoother

Plane by plane, keep a best estimate $\mathbf{s}$ and its covariance C.

Predict: $\mathbf{s}_{k|k-1} = f(\mathbf{s}_{k-1}), \quad C_{k|k-1} = F\,(C_{k-1} + Q)\,F^\top.$

Update: with $H = (1, 0, 0)$ (we measure x only),

$$K = C_{k|k-1} H^\top \big(H C_{k|k-1} H^\top + \sigma^2\big)^{-1}, \quad \mathbf{s}_k = \mathbf{s}_{k|k-1} + K\big(m_k - H\mathbf{s}_{k|k-1}\big), \quad C_k = (I - KH)\,C_{k|k-1}.$$

After the last plane the RTS smoother runs backwards so that every plane benefits from all hits. Cost $O(N)$. With the right Q, Kalman equals the full global fit.

Starting point. Proton mode: q/p from the known beam energy, with an uncertainty matching its $\sim 1\%$ spread. Di-muon mode: no prior on p; start from a straight-line fit of planes 1–6 and a large uncertainty on q/p, then iterate.

Three known traps

1. Scattering happens at the previous plane and is then propagated: $F(C + Q)F^\top$, not $FCF^\top + Q$.
2. The χ^2 uses predicted residuals (before the update), not smoothed ones.
3. In the magnet the linearisation depends on the track. Iterate: refit around the previous solution until it stops changing.

3.5 Vertex fit and invariant mass (di-muon mode)

Each muon is first fitted alone (Kalman + smoother). We then extrapolate both tracks to the vertex region and look for the common point.

Counting degrees of freedom. This is where 2D matters:

Case	unknowns of the vertex	constraints (1 per track)	ndf
Prompt, $z_v = z_T$ known	x_v	2	**1**
Displaced, z_v free	x_v, z_v	2	**0**
Displaced + beam-spot constraint on x_v	x_v, z_v	2 + 1	**1**

Two non-parallel lines in a plane always cross. With ndf $= 0$ the "vertex" is just that crossing point: the fit cannot correct the tracks and the mass does not improve. With ndf ≥ 1 the constraint pulls both tracks towards a common origin, and their directions (hence the mass) improve.

The fit (prompt case). Unknowns: x_v and, for each track, $(t_{x,i}, (q/p)_i)$ at the vertex: 5 numbers. Measurements: each track's fitted state $\hat{\mathbf{q}}_i$ at z_T with covariance C_i: 6 numbers. Minimise

$$\chi^2_{\text{vtx}} = \sum_{i=1,2} \big(\hat{\mathbf{q}}_i - \mathbf{q}_i\big)^\top C_i^{-1} \big(\hat{\mathbf{q}}_i - \mathbf{q}_i\big), \qquad \mathbf{q}_i = \big(x_v,\, t_{x,i},\, (q/p)_i\big),$$

which is linear, so the solution is closed form. χ^2_{vtx} must follow a χ^2 with 1 degree of freedom.

Invariant mass. With $p_i = |1/(q/p)_i|$, $\theta_i = \arctan t_{x,i}$, $E_i = \sqrt{p_i^2 + m_\mu^2}$:

$$m_{\mu\mu}^2 = 2m_\mu^2 + 2\big(E_1 E_2 - p_1 p_2 \cos(\theta_1 - \theta_2)\big).$$

Its error σ_m follows from the Jacobian of m with respect to the four track parameters and their covariance. Check it with the pull $(m - M)/\sigma_m$.

Comparison: the mass peak reconstructed with (i) naive global fit, (ii) Kalman, (iii) Kalman + prompt vertex constraint; and prompt vs. displaced vertex.

3.6 Optional method 3: Bayesian fit with MCMC (Metropolis–Hastings)

Treat the initial state and the scattering angle in each phantom slice as unknowns and sample their posterior. Slow, but makes no Gaussian assumption: an independent check of the Kalman errors, and a method that still works with non-Gaussian tails (S-c), where Kalman struggles.

4 Machine learning, compared fairly

ML is used as *one more method* in the comparison tables, evaluated with exactly the same tests as the classical methods (pulls, χ^2, resolution, CPU time). We pick **one** of ML-1/ML-2 for the core extension, and ML-3 only if time allows.

Label	Idea	Compared with	Key question
ML-1	PINN propagator	RK4, RK45 (R-b, R-c)	A network $\mathbf{s}(z; \mathbf{s}_0)$ trained so that it satisfies (1) (residual of the ODE in the loss), for a range of initial states. Is it accurate enough? Is it faster when propagating many tracks at once? The Jacobian comes free from autodiff.
ML-2	Surrogate propagator	RK45 in the M-c field	A network learns the map "state at magnet entrance → state at exit" from RK45 samples, then replaces RK45 inside the Kalman filter. Do the pulls stay $\mathcal{N}(0, 1)$?
ML-3	Direct regression	Kalman	A network predicts p (or $m_{\mu\mu}$) directly from the hits, with an uncertainty (Gaussian likelihood output or an ensemble). It may match the value; does it give a trustworthy error?

Rules for the ML part

Separate training and test samples generated with different seeds. The network never sees the test truth. Report the training cost too, not only the inference time. A PINN for the magnetic field itself is *not* used: in strict 2D only $B_y(x, z)$ at $y = 0$ appears, and Maxwell's equations do not constrain it without the y dependence.

5 How we compare models

One reference configuration, one change at a time. The reference is: M-a, P-a, E-b, S-a, R-b, Kalman. Each study changes *one* ingredient and keeps the rest fixed. Every study is one small YAML file in `config/studies/` and one figure.

Study	What changes	Main figure
Field	M-a → M-b → M-c → M-d	momentum bias vs. magnet model
Propagator	R-a . . . R-f	error vs. field evaluations; CPU time
Phantom	P-a → P-b → P-c	WEPL and its resolution; WEPL(x) profile
Energy loss	E-a → E-b → E-c	range; σ_p/p at exit
Scattering	S-a → S-b → S-c	pull widths and tails
Fit method	naive, global, Kalman, (MCMC), (ML)	resolution and CPU time vs. number of planes
Vertex	none / prompt / displaced	mass peak width

The mismatch study. Generate the data with one model and fit with another (for example, truth E-c + S-c, filter E-b + S-a). This answers a real question: *how wrong can the model inside the fit be before the result becomes biased or the errors become dishonest?* Present it as a small matrix of pull widths (rows: truth model, columns: filter model). The diagonal must be ≈ 1.

6 Validation: how we know it works

Test	What we expect
Pulls	$(\hat{s} - s_{\text{true}})/\sigma_{\hat{s}}$ is $\mathcal{N}(0,1)$ for x, t_x, q/p. Width > 1: errors underestimated. Offset: bias.
Track χ^2	χ^2 distribution with $N_{\text{hits}} - 3$ degrees of freedom.
Momentum resolution	σ_p/p vs. p: scattering dominates at low p, hit resolution at high p. Compare with Gluckstern's formula.
WEPL	σ_{WEPL} vs. phantom thickness; WEPL(x) for P-c.
Range	integrated Bethe–Bloch range vs. NIST PSTAR.
Vertex	pull of x_v is $\mathcal{N}(0,1)$; χ^2_{vtx} has mean 1 (prompt case).
Mass	peak centred at M; mass pull $(m - M)/\sigma_m$ is $\mathcal{N}(0,1)$.
Negative control	deliberately wrong Q: the pulls must get wider. If not, the test is not sensitive.

Unit tests per module (a module is done when its tests pass, not when it runs):

Module	Acceptance test
Field	M-c tends to $\mu_0 K$ for infinite plates; $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{B} = 0$; M-b tends to M-a; M-d interpolation error decreases with grid spacing.
Propagators	R-b error $\propto h^4$; all agree with R-e in uniform field; forward then backward returns the start; R-c respects its tolerance.
Materials	layered phantom stops at every interface; WEPL of P-a equals its thickness.
Simulation	fixed seed reproduces; hit residual width equals σ; scattering width equals the chosen model.
J/ψ generator	mass computed from the *true* muons equals M to machine precision; both muons forward ($p_z > 0$).
Kalman	with $B = 0$ and no scattering it equals ordinary least squares; innovations are white noise.
Vertex fit	with zero noise it returns the true vertex; χ^2_{vtx} has mean 1.

7 Code organisation

One Python package, one job per module, a **registry** of models: each model registers itself under a name and the config chooses by name. Adding a new model never touches the fitting code.

```
phy571-tracking/
|-- README.md                 how to install and run; which script makes which figure
|-- requirements.txt          numpy, scipy, matplotlib, pyyaml, pytest (+ torch, ML only)
|-- config/
|   |-- proton.yaml           reference configuration, proton mode
|   |-- dimuon.yaml           reference configuration, di-muon mode
```

```
|   `-- studies/              one small YAML per comparison (overrides one key)
|-- src/ptrack/
|   |-- registry.py           register / look up models by name
|   |-- geometry.py           planes, phantom, magnet, target positions
|   |-- field/                hard_edge.py, enge.py, biot_savart.py, fieldmap.py
|   |-- materials/            tables.py (rho, Z/A, I, X0), phantom.py (layers, insert)
|   |-- physics/              eloss.py (E-a, E-b, E-c), scattering.py (S-a, S-b, S-c)
|   |-- propagators/          euler.py, rk4.py, rk45.py, rkn.py, circle.py, jacobian.py
|   |-- sources/              proton_beam.py, jpsi.py
|   |-- simulation.py         propagate + random kicks -> true states
|   |-- detector.py           smear true x into hits
|   |-- fitting/              global_fit.py, kalman.py, vertex.py, mass.py, mcmc.py
|   |-- ml/                   pinn_propagator.py, surrogate.py, regression.py
|   |-- validation.py         pulls, chi2, resolutions, WEPL, mass
|   `-- plotting.py           all figure styles in one place
|-- tests/                    one test file per module (see the table above)
|-- scripts/
|   |-- run_simulation.py     config -> data/sim_*.npz
|   |-- run_reconstruction.py config -> data/fit_*.npz
|   |-- run_study.py          runs one study YAML end to end
|   `-- make_figures.py       ALL report figures from data/ -> figures/
|-- notebooks/                quick exploration only
|-- data/  figures/           generated, not committed
`-- report/                   LaTeX sources of the report and slides
```

Rules of the game.

- Every number lives in the config. No magic constants. Each output file saves the config and seed it was made with.
- Code flows one way: field, materials, physics → propagators → simulation / fitting → validation. Nothing imports backwards.
- Simulation and filter use the same propagator code; the simulation only adds random kicks.
- `ml/` is the only place that imports PyTorch; the rest of the project runs without it.
- Run `pytest` before every merge. Units: lengths in mm, momentum in GeV/c, field in T, energy in MeV, written at the top of each file.

8 Work plan

8.1 Who does what

	Person A: transport	**Person B: estimation**	**Person C: physics & scenarios**
Owns	`field/`, `propagators/`, `ml/` (ML-1 or ML-2)	`fitting/` (global, Kalman, RTS, vertex, mass, MCMC)	`materials/`, `physics/`, `sources/`, `simulation.py`, `detector.py`
Numerics	ODEs, RK family, convergence, Biot–Savart integration, interpolation, neural networks	linear algebra, nonlinear filtering, constrained fits, sampling	Monte Carlo, special relativity (boosts), material models, WEPL
Own tests	exact circle, h^4, forward–backward, $\nabla \cdot \mathbf{B}$	$B = 0$ vs. least squares, pulls, χ^2, vertex	range vs. NIST, Highland width, J/ψ mass at truth level
Reviews	that numerical and analytic Jacobians agree (B's code)	that Q matches the simulation physics (C's code)	that the propagators handle layer boundaries (A's code)

Together: the comparison studies, validation, report and presentation. Short weekly meeting; each milestone reviewed by another person. Total budget $\approx 3 \times 54 \approx 160$ h.

8.2 Six weeks

Week	Work	Milestone (done means)
1	Repo, config, registry, conventions. Straight tracks, linear Kalman and least squares (B). Hard-edge field + exact circle (A). Water material, Bethe–Bloch, range (C).	Kalman = least squares on straight tracks; range agrees with NIST.
2	RK4, Euler, analytic circle, convergence study (A). Hits, Kalman with analytic Jacobian, global fits (B). Highland, phantom slices, layered phantom P-b (C).	h^4 plot; exact circle reproduced; simulated scattering width = Highland.
3	Enge and Biot–Savart fields, RK45, RKN (A). Numerical Jacobian, iterated filter (B). J/ψ generator, di-muon geometry, sanity checks (C).	Kalman converges in the realistic field; truth-level J/ψ mass = M.
4	Field map + interpolation; propagator comparison (A). RTS smoother, phantom in Q, vertex fit and mass (B). P-c insert, E-c straggling, S-b/S-c (C).	Pulls $\approx \mathcal{N}(0,1)$ with phantom; first mass peak.
5	All: run every study YAML, mismatch matrix, global vs. Kalman (accuracy and CPU), prompt vs. displaced vertex.	All core figures produced by `make_figures.py`.
6	One extension (ML-1 or ML-2 by A; MCMC by B if time), report, slides. No new core code.	Reproducible submission.

8.3 Extensions (only if the core is validated)

Prio.	Extension	What it adds	Effort
1	ML-1 or ML-2	ML propagator compared with RK4/RK45 in accuracy, speed, and pulls	10–15 h
2	MCMC fit	third fitting method, checks Kalman errors, works with S-c tails	8–12 h
3	ML-3 regression	does a network give honest errors?	6–10 h
4	Combinatorial background	random $\mu^+\mu^-$ pairs under the J/ψ peak; signal extraction	4–6 h
5	Fokker–Planck with Crank–Nicolson	beam envelope from a PDE, cross-check of the Monte Carlo	5–8 h

8.4 If things go wrong

- Iterated Kalman does not converge ⇒ back to uniform field and linear filter, then add complexity one step at a time.
- Bethe–Bloch is a pain ⇒ constant $\mathrm{d}E/\mathrm{d}z$ (E-a). The qualitative effect survives.
- Biot–Savart is a pain ⇒ the Enge model (M-b) already gives a fringe field.
- The di-muon mode is late ⇒ keep only the prompt case with the vertex constraint; drop the displaced variant.
- Kalman and global fit give the same answer ⇒ that is a result: report the regime where they agree and why.
- Running out of time ⇒ drop extensions (ML included), never validation or the report.

9 Expected results

1. A reproducible simulator with interchangeable models; every figure regenerated by one script.
2. Propagator comparison: convergence (h^4), error vs. field evaluations and CPU time for R-a to R-e (and R-f).
3. Magnet comparison: momentum bias from ignoring or approximating the fringe field.
4. Correct pulls and χ^2 for x, t_x, q/p; the mismatch matrix.
5. Momentum resolution vs. p compared with Gluckstern; WEPL resolution and WEPL(x) for the layered and insert phantoms.
6. Quantitative Kalman vs. global fit comparison (accuracy and CPU time).
7. J/ψ mass peak for each method, vertex resolution, and the effect of the vertex constraint (prompt vs. displaced).
8. One extension, ideally an ML propagator compared fairly with RK4/RK45.

The conclusion is not "Kalman is always better" or "ML is faster", but a quantitative map of where each model and method works and where it breaks down. A simple, deeply validated model beats a large unchecked one.

10 Link with the course

- **Central:** Monte Carlo simulation (Lecture 2); ODE integration and convergence studies, now with several integrators (Lecture 1); numerical linear algebra (the filter and the vertex fit); numerical integration (Biot–Savart, range); interpolation (field map).
- **Through extensions:** Metropolis–Hastings (Lecture 2) for the MCMC fit; Crank–Nicolson (Lecture 3) for Fokker–Planck.

Course methods are used where they improve the project, never forced in.

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